Temur Makharadze

Mechanical Engineering Student, B.Sc. 2027

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Final-year mechanical engineering student with industrial experience in CNC machining and scan-based reverse engineering (Nasolutions, since 02/2026), SOLIDWORKS CSWP certification, and a track record of building simulation, embedded-control and competition robotics projects. Accepted for a research internship at Forschungszentrum Jülich in February 2027. Seeking a master's in mechanical engineering in Germany, with motorsport as the long-term goal.

EDUCATION

B.Sc. Mechanical Engineering, Agricultural University of Georgia, Tbilisi — 09/2023 – summer 2027 (expected)

- Coursework: Finite Element Method, Machine Elements, Strength of Materials, Theory of Mechanisms and Machines, Automotive Theory, Thermodynamics and Heat Transfer, Mechatronics, Automatic and Digital Control, Manufacturing Processes, CAD Methods.
- 240 ECTS, four-year programme taught in Georgian, with three engineering course projects (sophomore, junior, senior).

EXPERIENCE

Research Intern (accepted), Forschungszentrum Jülich, Institute of Technology and Engineering (ITE) — starts 02/2027
Helmholtz Association, Jülich, Germany

- Selected from about 100 students by the institute director, Prof. Dr. Ghaleb Natour.
- Preferred group: Simulation and Calculation (FEA, thermal and flow analysis of components under extreme loads).

Engineering Intern, Nasolutions (manufacturing company), Georgia — 02/2026 – present

- Machine aluminium, stainless steel, carbon steel and titanium parts on 3-axis CNC mills; run CNC routers, laser cutters and 3D printers.
- Reverse-engineer mechanical and automotive parts from 3D scans into SolidWorks models.
- Co-designed a 6-axis industrial robot arm for welding. Use SolidWorks daily for part and assembly design.

PROJECTS

Sphere-1: free-flying inspection drone for low Earth orbit (thesis project) · portfolio page — 2026

- Designed a 23 cm, 500 g vehicle with eight cold-gas CO2 nozzles: propellant selection, nozzle sizing against the CO2 triple point, and an actuator layout chosen on valve mass.
- 10 g of propellant holds three-axis pointing for 3.3 days; a re-planned survey cut propellant use by two thirds.

Sphere-1 flight control, hardware in the loop · portfolio page — 2026

- 6-DOF rigid-body simulation in Python with a thruster control law and allocation across eight nozzles, shown in a live 3D view with delta-V and propellant readouts.
- ESP32 flight controller at 200 Hz with a real IMU; gyro feedforward cut tracking lag from 11° to 1.2°.

Competition robots · sumo robot · football robot — 2024 – present

- **1st place**, university sumo championship (2 kg class, 03/2024) · **3rd place**, national sumo championship (2 kg, 07/2024) · **4th place**, national BattleBots championship (15 kg class, 07/2024).
- Designed and built a 3 kg international-class sumo robot alone: aluminium chassis CNC-machined from my own G-code, 6.44:1 gearbox of hardened stainless gears, magnets for traction.
- Mechanical and electrical design of a 3.9 kg omni-wheel football robot (team of two); cut mass with topology studies.
- Now designing and building 1.5 kg and 5.4 kg combat robots for Robo Battle 2026 (late November).

LEADERSHIP

Competition designer, university engineering championship, Agricultural University of Georgia

- Designing an autonomous RC car race on a NASCAR-style oval for first-year students, covering mechanical, electrical and software work. Set the technical scope, rules and difficulty.

SKILLS

CAD	SolidWorks (daily industrial use), AutoCAD, reverse engineering from 3D scans
Manufacturing	3-axis CNC milling, CNC routing, laser cutting, 3D printing, 3D scanning
Simulation, code	ANSYS Mechanical (static, transient and modal FEA), Python (6-DOF dynamics, control)
Embedded	ESP32, IMU sensors, serial communication, PWM control
Certifications	SOLIDWORKS CSWP, CSWA · "Modern Methods in Engineering" block course (FZ Jülich ITE, 09/2026)
Languages	Georgian (native), English (fluent; IELTS Academic planned for 11/2026), German (A2)
Interests	CFD, FEA, vehicle dynamics, mechatronics, motorsport and Formula Student